Forced and Free Content in the GeoVac Framework: An Observations Paper

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Abstract

The GeoVac framework produces a wide class of structural facts (the (n, l, m) label set and its SO(4) Casimir spectrum, S^3 outer manifold, master Mellin engine, inner-algebra factor count, the D_F moduli dimension $\dim \mathcal{M}(D_F) = 32$ at one generation (matter-sector projection rank 16; the factor count and this dimension hold relative to the Hopf-tower organizing reading, the adopted Upgrade B axiom and the imported CCM representation), two-term-exact gravity, and many more), admits the gauge group $U(1) \times SU(2) \times SU(3)$ under an organizing reading plus an adopted axiom (Paper 32; not a derivation), and consumes specific numerical content as external input (Yukawa values, generation count, calibration cutoff moments, atomic-physics screening exponents, multi-loop QED counterterms, etc.). We catalogue both sides of this *forced/free seam* across the corpus — 44 forced entries (the gauge-group entries forced relative to the adopted Upgrade B axiom), 4 Observations (κ, matched not forced; two released-axis results that lack a derivation, the decompactification front and the four-body r_{12} bridge; and the rung rule, a pattern across the tested instances), 1 admitted-not-forced entry, 27 calibration entries and 3 walls, spanning foundations, gauge structure, periods, gravity, QED, the Standard Model inner factor, multi-focal-composition walls, chemistry / quantum computing scaling, foundational calibration and the released axes of the chemistry program: a second nucleus, the decompactification front, a third and a fourth centre, bending, a three-electron chain, the four-body r_{12} bridge, four linked densities and a ring of three densities, and, from the label set itself, a spin grown large, the hydrogen continuum, a finite nucleus and its magnetization, and the Kustaanheimo–Stiefel lift, each recorded with the number class that enters when the axis is released, where one is assigned — weight-one logarithms at two centres, an elliptic period at three, a K3 holomorphic period for the chain, a period not built from elliptic ones for four linked densities, and, at the fibre, an elliptic period again for a Coulomb ring of three, with dilogarithms at one centre; $\pi^{1/2}$ and e^{-x^2}, only as limits, for the spin; and one exponential of π for the continuum; while a finite nucleus consumes moments of its charge as matching coefficients. We identify a cross-domain unity pattern (the same wall appears in seven settings, all carrying the same multi-focal-composition signature, the two nuclear ones sharing the textbook matching mechanism of nonrelativistic QED) and test five candidate principles for what discriminates the two sides: multi-focal depth, period-content classifiability, dimensional character, compactness inheritance, and packing-reachability. The fifth candidate (packing-reachability: an entry is forced iff a witness structural derivation exists from {Paper 0 + standard real-spectral-triple axioms + Hopf-tower truncation + Bertrand + Upgrade B + CCM canonical SM representation}) *consolidates* the forced/free boundary rather than independently discriminating it: its 98.3% agreement with the F/C status (59/60, on the 2026-06-08 labelling in which κ was counted forced) is an internal-consistency check confirming the catalogue applies its conventions uniformly — the packing-reachable tag is the F/C label by construction — not an independent accuracy. The independent discriminator signal is carried by multi-focal depth, period class, and the two-family decomposition. The earlier-named two-family decomposition (multi-focal composition vs. inner-factor input data) is preserved as the structural failure-mode story beneath the principle: both families share the same surface verdict (packing-unreachability) but exhibit different structural reasons for it. This paper consolidates what was previously scattered across CLAUDE.md §1.7 working hypotheses, dead-end records, and individual sprint memos.

1 Introduction: the seam and why naming it matters

GeoVac is a discretization framework built on the label set (n, l, m) of the unit-three-sphere harmonics, whose per-shell angular count $2l + 1$ the Paper 0 packing reproduces [1], together with Fock's conformal equivalence (whose chain Paper 7 verifies) between the continuum S^3 Laplace–Beltrami operator and the Schrödinger problem with $-Z/r$ potential [2]. (The graph's edge Laplacian is *not* that operator and does not converge to it: it preserves l and splits into $n_{\max}$ disconnected blocks, Paper 0 §VI.) From this minimal foundation the framework has been shown to generate a large class of structural facts: the angular sparsity theorem of Paper 22 [5], the chemistry Pauli count of Paper 14 [3] (exactly linear in the qubit count across molecules at fixed basis, $N_{\text{Pauli}} = 27.90 \times Q$), the spectral triple structure of Paper 32 [11] with state-space Gromov–Hausdorff convergence (proved) of the true-window truncations to the round-S^3 triple for the commutative factor (Paper 38 [15], main theorem: a Fejér-kernel bound $(4/\pi) \log n/n$, **[INTERNAL THEOREM + MEASURED]**, and a measured $O(1/n)$ with a tapered amplitude), the master Mellin engine's tagging of the transcendentals in the census of GeoVac spectral data (a census and a reading, not a classification; Paper 18 §III.7; Paper 32 §VIII case-exhaustion theorem, **[CONDITIONAL]** and scoped to the first fifteen projections [4, 11]), and the Forced-Count Theorem (moduli dimension $\dim \mathcal{M}(D_F) = 32$ at one generation, matter-sector projection rank 16) (Paper 32 §VIII, `thm:forced_count`). The gauge group $U(1) \times SU(2) \times SU(3)$ is catalogued as forced only *relative to the adopted Upgrade B axiom*: it follows from the Bertrand × Hopf-tower organizing reading plus that axiom (Paper 32 §VIII.B), not from a derivation.

The framework also consumes specific numerical content that it does not autonomously generate. The Yukawa values cannot be selected by the inner-factor structure (Sprint H1 non-selection theorem, empirically confirmed at 162 PSLQ cells against the inner-factor $M_1 \cup M_2$ pure-Tate basis [11, 32]). The generation count $N_{\text{gen}} = 3$ is packing-unreachable in the current axiom system (Direction 2 NO-GO, Read 2 NO-GO [34]). The chemistry-side inner-region correlation wall (*W1e*) consumes external atomic-physics screening data (Clementi–Raimondi exponents) and cannot be closed by any framework-internal rational/algebraic modification, as verified by six independent sprint attempts (CLAUDE.md §3 entries F1–F6 plus Schmidt, core correlation, and basis enlargement). The multi-loop QED renormalization counterterms (Z_2, δm) are likewise consumed rather than generated (LS-8a wall [14]).

The line between what the framework forces and what it consumes is itself a structural property of the framework. It has been repeatedly named — the *structural-skeleton scope* reading of CLAUDE.md §1.7 WH3, the *multi-focal-composition wall pattern* crystallized at six instances, the *external-input three-class partition* (Calibration / Multi-focal composition / Multi-determinant correlation) — but never consolidated. Each naming has produced a clearer statement than the previous one, but the underlying inventory of which physical inputs sit on each side, and which discriminator (if any) governs the partition, has been distributed across many sprint memos, dead-end rows, and working-hypothesis bullets.

This paper consolidates the inventory and tests, at preliminary level, five candidate discriminator principles against it. The goal is not to close the seam — we do not yet claim to have identified *the* principle that decides which side a given physical input falls on — but to name the question sharply enough that it admits a determinate answer.

Why an observations paper, not a theorem paper. The format follows Paper 2, Paper 25, and Paper 33 (the existing observation-genre papers in the GeoVac corpus). We describe a structural pattern that the framework exhibits across multiple independent domains, we cite the underlying theorems and dead-end records that establish each entry, and we probe candidate discriminators without commitment. Where a finding has theorem-grade status it is named as such; where it is a numerical / empirical observation only, the honest scope is recorded explicitly. The companion catalogue `docs/forced_free_seam.md` carries the full table; this paper is the

publication-grade framing.

What we promise to deliver in this paper.

- §2 — forced-side inventory across nine domains (foundations, gauge, periods, gravity, QED, Standard Model inner factor, chemistry/QC scaling, foundational calibration, and — added 2026-10-15 — the released axes of the chemistry program, §2.7; the load-bearing sub-domains).
- §3 — free-side inventory with non-selection theorems and documented negative searches.
- §4 — the cross-domain unity: the multi-focal-composition wall pattern at seven independent instances.
- §5 — formal probe of five candidate discriminators (Sprint C2 finding: packing-reachability is the consolidated F/C boundary; its 98.3% agreement with the F/C status is an internal-consistency check, not an independent discriminator).
- §6 — open questions; the "second packing axiom" status.

What we do NOT promise.

- A theorem stating "GeoVac forces X iff X has property P." No such theorem is proved or even decisively favoured in this paper.
- A derivation of any presently-free quantity from a single GeoVac principle.
- A demonstration that the present seam is the only possible seam (a "second packing axiom" could shift the line and remains the named long-range frontier).

2 The forced side

We organize the forced inventory by sub-domain. Each entry carries a mechanism (what does the forcing), a witness (theorem cite or bit-exact verification), and three discriminator columns: multi-focal depth (MF), period class in the master Mellin engine (P; values M_1, M_2, M_3, "outside," "none," "meta" for the engine itself, "N/A" for non-period observables, and — for the released-axis rows of §2.7 — "weight-1," "elliptic," "K3," "CY-type," classes that the Mellin engine does not reach and that leave the polylogarithm weight tower), and dimensional character (Dim; dimensionless ratio vs. dimensionful scale). Sub-domains are referenced by the letter codes A–J used in the companion catalogue `docs/forced_free_seam.md`. A fifth status, W (wall), was added to the catalogue 2026-10-15 for a proven obstruction with its mechanism pinned — the third verdict of the project's internal mission statement alongside forced and free; the walls' working record is `docs/walls/register.md`, and the door-indexed view of every released axis with its frozen falsifier is `docs/atlas.md`.

2.1 Foundations / skeleton (A)

The packing construction of Paper 0 motivates the labels (l, m) (with n and s recorded there as Observations); through Fock's projection, the continuum S^3 Laplace–Beltrami operator acts on the label set (n, l, m) exactly as the diagonal $SO(4)$ Casimir, with spectrum $-(n^2-1)$, at every cutoff (the graph's edge Laplacian does *not* converge to it; it splits into $n_{\max}$ disconnected l-blocks) [1, 2]. Paper 7's eighteen symbolic checks (ten limiting-case, definitional or non-discriminating) of Fock's S^3 conformal chain [2] pin the outer manifold; Bertrand's classical

theorem [25] combined with SO(4) closure forces it to be S^3 rather than a different homogeneous space. The matched energy scale $\kappa = -1/16$ is exact rational, but it is matched rather than forced: it maps the graph's $\lambda_{\max} \to 8$ (twice the maximum degree, a grid band edge carrying no S^3 information) onto $-1/2$, and its coincidence with the Fock value $1/\Omega^4(0)$ is an Observation (Papers 0, 7). The shell degeneracy $g_n = n^2$ is representation-theoretic. The deeper organizing observation, established across Papers 18, 22, 24, 27, 32, and consolidated in the structural-skeleton scope pattern, is that the framework's discrete content lies in the closed/compact/Peter–Weyl regime; the continuum is what remains when compactness is released. (Compactness forces discreteness, A5; the converse fails — discrete spectrum is the compact-resolvent property, as for the Paper 24 oscillator on non-compact $\mathbb{R}^3$.) Two rows added 2026-10-15 make that pair explicit: A6 records the edge Laplacian $D - A$ as a wall (it preserves l, splits into $n_{\max}$ disconnected blocks with $\dim\ker L = n_{\max}$ at every cutoff, and does not converge to Δ_{S^3}; Paper 0 §VI, Paper 7, `tests/test_graph_s3_nonconvergence.py`), and A7 records the Bargmann–Segal lattice of the harmonic oscillator as forced π-free (bit-exact rational at every $N_{\max}$; Paper 24 [9]), the corpus's own instance of a discrete spectrum on a non-compact space, which is why compactness is recorded as sufficient for discreteness and not necessary.

2.2 Gauge structure (B)

The Standard Model gauge group is reached, under an organizing reading plus the adopted Upgrade B axiom (Paper 32; not a derivation), from two independent structural inputs: Bertrand's classical theorem (only $-Z/r$ and kr^2 central potentials produce closed orbits) and the Hopf-tower of spheres $S^1 \to S^3 \to S^5 \to \cdots$ truncated at $n \leq 3$ by the Hurwitz classification of associative real normed division algebras (Door 4d, summarized in [11], §VIII.B). This produces

$$\mathcal{A}_F \;=\; \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}), \tag{1}$$

with $\mathbb{C}$ at $n = 1$ from the unique division algebra of S^1, $M_3(\mathbb{C})$ at $n = 3$ from the Hurwitz fallback at S^5, and $\mathbb{H}$ at $n = 2$ *admitted* but not autonomously forced over $M_2(\mathbb{C})$ in the J-sign-table audit (Door 4c). The admitted-not-forced status of $\mathbb{H}$ is the lone admitted entry in our catalogue and is recorded as B5.

The Forced-Count Theorem [11] (Paper 32 §VIII, `thm:forced_count`), given the admitted $\mathcal{A}_F$ and the imported CCM representation, chains the standard real-spectral-triple axioms (Hermiticity, KO-dim-6 chirality, J-reality, order-one) into the bit-exact moduli dimension

$$\dim_{\mathbb{R}} \mathcal{M}(D_F) \;=\; 32 \quad \text{(one generation; matter-sector projection rank 16)}, \tag{2}$$

(Under a linear $*$-representation of $\mathcal{A}_F$ the full-axiom D_F moduli dimension is 32 with matter-sector projection rank 16, Paper 32 `thm:forced_count`.) The eight free parameters per generation in the diagonal slice are exactly the Yukawa values (charged-lepton + quark sector). The theorem forces the *count*; it does not force the *values*.

The Higgs admission — the structural fact that the Mexican-hat field-theory structure is allowed within the inner factor's spectral action — is the conclusion of Sprint H1 (POSITIVE-THIN verdict, Paper 32 §VIII.C). It is an admission of *structure* (given an imposed Yukawa), paired with a non-forcing of *value* via Sprint H1's Yukawa non-selection theorem.

2.3 Spectral / period content (C)

The master Mellin engine of Paper 18 §III.7 [4], sharpened in Paper 32 §VIII via the case-exhaustion theorem [11], tags the π-sources in chains of the first fifteen Paper 34 projections (**[CONDITIONAL]**) by three sub-mechanisms, M_1 (volume slot), M_2 (Seeley–DeWitt heat kernel) and M_3 (vertex-parity Hurwitz). The engine's defining transform, slot assignments and period-ring homes are stated once, in Paper 18 §III.7 (`sec:mellin`), with Paper 55 as

the census; this paper does not restate the table (the rows C1–C6 of the companion catalogue point at those two owners). The ring inclusions are theorem-grade (Paper 55); their exhaustiveness is the conditional case-exhaustion theorem plus a census. The signature prefactor $M_1 = \mathrm{Vol}(S^2)/4 = \pi$ recurs across the corpus; the Fejér-moment constant $4/\pi$ of the van Suijlekom state-space Gromov–Hausdorff bound (a kernel constant, not the convergence rate) is read (a reading, not a derivation) as an M1 value (dual-Coxeter normalisation; $2/\pi$ on the unit S^3) derived at rank 1; whether it is the same for every compact connected simple Lie group is open (Paper 38, Paper 40 [15, 16]; the rank-≥ 2 extractions compare mismatched conventions).

2.4 Gravity (D)

The spectral action on S^3 is two-term-exact by the Bernoulli identity $\zeta_{\mathrm{unit}}(-k) = 0$ for all $k \geq 0$ [19]; the closed forms for S_{BH}, G_N and $s_{\min}$, the Sommerfeld–Cheeger cone coefficient $-1/12$ and the replica derivative $+1/6$ are Paper 51's (§G4) and are not restated here; the catalogue rows D1–D4 point at them. The relations among G_N, Λ_{cc}, and S_{BH} are forced by Wald's theorem (two-term-exactness $\Rightarrow$ pure Einstein action $\Rightarrow$ action-$G \equiv$ entropy-G). The relations are forced; the cutoff function f itself (its $\varphi(0), \varphi(1), \varphi(2)$ moments) is not — see §3.

2.5 QED and the α coincidence (E)

Paper 28 establishes five theorems on S^3 QED [10]: the self-energy structural zero $\Sigma(n_{\mathrm{ext}} = 0) = 0$, the squared-Dirac period ring $\zeta_{D^2}(s) \in \bigoplus_k \pi^{2k} \cdot \mathbb{Q}$ at integer s (T9), the χ_{-4} identity $D_{\mathrm{even}}(s) - D_{\mathrm{odd}}(s) = 2^{s-1}(\beta(s) - \beta(s-2))$, the $\zeta(3)$ complementarity, and the product survival rule. The Paper 33 vertex-promotion analysis recovers the full 1+6+1 partition of QED selection rules at the structural level [12].

The fine-structure-constant coincidence is decomposed into three spectral homes of distinct type: $B = 42$ (Casimir trace), $F = \pi^2/6$ (Fock-degeneracy Dirichlet at the packing exponent), $\Delta = 1/40 = 1/g_3^{\mathrm{Dirac}}$ (inverse Dirac mode degeneracy). Each has an identified spectral home; why these three ingredients are selected is open (Paper 2). Twelve mechanisms have been eliminated for the combination rule $K = \pi(B + F - \Delta)$ as a single-principle derivation; the combination rule itself remains an Observation and is recorded as a calibration entry in §3.

2.6 Chemistry / quantum computing scaling (H)

Paper 22's potential-independent angular-sparsity theorem [5] forces 6.06% ERI density at $l_{\max} = 3$ regardless of the potential. Paper 14's universal $N_{\mathrm{Pauli}} = 27.90 \times Q$ holds across the main-group composed molecules of the library ($30.03 \times Q$ for the ten d-block hydrides) with the isostructural-invariance corollary (CO $=$ N_2 $=$ F_2 $=$ 2,790 at $Q = 100$) [3]. The Pauli ratio balanced/composed grows as $O(B^2)$ at the structural-architectural level. Paper 23's Fock projection rigidity theorem [6] establishes that the S^3 conformal projection is unique to $-Z/r$ and does not transfer to the harmonic oscillator or Woods–Saxon, with downstream consequence that the harmonic-oscillator shell-closure numbers $\{2, 8, 20, 40, 70, 112\}$ (the spin-orbit-free closures, distinct from the physical magic numbers $\{2, 8, 20, 28, 50, 82, 126\}$) are forced by shell counting on the Bargmann–Segal lattice (shell degeneracy $(N+1)(N+2)$ with spin, cumulative $(N+1)(N+2)(N+3)/3$; Paper 24) — a property of the HO lattice, independent of the Fock rigidity result.

2.7 Released axes: the decompactification doors (J)

Sub-domain J (sixteen rows) records the released axes, eleven from the chemistry program and five from the label set itself (J12–J16): each row is one label of the compact skeleton released toward the continuum, with the number class that enters once it is released. An F on a J row is relative to the release: it covers what the release leaves standing, such as the m rule once l is

released across centres, and the number class of what the release lets in, fixed by the released integral itself with no calibration input. The witness is a symbolic derivation; high-precision quadrature and exact searches are checks, and a part that rests on measurement alone is named in its row and is outside the F. These rows are therefore not packing-reachable in the sense of P5 (§5.5), which is why P5's agreement is quoted on the 60-entry catalogue. The rows are the first place the corpus indexes together a ladder it had stated only piecewise (Paper 18 §III, Papers 59 and 63): at one centre the two-electron classes are exact rationals (Paper 58) and the one-centre limit of the three-density chain below is rational, while the ring of J11 carries dilogarithms there; at two centres the seed set is $\{e^a E_1(a), \ln, \gamma_E\}$, weight one and π-free (Paper 18); at three centres the fibre is a genus-one curve and the class leaves the polylogarithm tower for the elliptic one (Paper 59); a fourth centre adds nothing new; a three-electron chain over two-centre densities with two momentum integrations lives on a K3 surface whose holomorphic period is a sum of two products of elliptic periods; four linked densities give the first period not built from elliptic ones; and a Coulomb ring of three densities does not climb, sitting at the fibre on the elliptic rung below the chain, while the correlated triangle with exponential geminals, whose loop has no uniqueness step, sits with the chain at $\gamma = 1$.

- **J1, a second nucleus (F).** The prolate diatomic releases l across centres while m survives as additive $U(1)$ conservation (Paper 58's abelian-residue theorem, `thm:abelian_residue` [21]; the m rule is Curie's and Bethe's). The two-centre repulsion integrals close in weight one: (AA|BB) in elementary form, decided π-free by Lindemann–Weierstrass; (AA|AB), (AB|BB) in $\{E_1, \ln\}$; (AB|AB) in $\{E_1, \ln, \gamma_E\}$ (Paper 12's closed-form seeds [20]; Neumann, Roothaan, Ruedenberg). The seed classes are *counted*; only the elementary block is *decided* — weight one is not the same as decidable, and the catalogue keeps the distinction.

- **J2, the decompactification front (Obs).** The coupling that l forbade turns on continuously in R, with the front set by the decay length n/Z and the label staying Boolean (Paper 58, measured front and exponent; the rate bound $0 < dA/dc^2 < 1$ is a DLMF spheroidal identity). No new number class enters; Herring's asymptotics is the prior art.

- **J3, a third nucleus with a shared centre (W).** The (XY|XZ) integral's fibre is the genus-one curve $y^2 = (c_1k^2 + 1)(c_2k^2 + 1)$ with period $K(m)$ (Paper 59 [22]); at the complex-multiplication fibre $\rho = 1/2$ the period is $\Gamma(1/4)^2/(4\sqrt{\pi})$ (Paper 61 [23]; Chowla–Selberg). The wall is on closed forms only: the radial core is a period of an irreducible connection, so it has no closed form through any factorization, and the elliptic-dilogarithm route is obstructed (Paper 59); whether the integrated integral has a finite closed form is open, and accurate evaluation is unaffected.

- **J4, a fourth distinct centre (F, pre-registered and confirmed).** The (XY|ZW) integral is still one radial k-integral with the same two dispersion roots, so no genus two appears at four centres (registered 2026-10-13 before the computation, confirmed against a McMurchie–Davidson evaluation; nearly a corollary of Paper 59). Its home is Paper 63 [24] (`sec:four_centre`), with the guard `tests/test_paper63_four_centre_genus.py`.

- **J5, bending (F for the order and labels).** Bending a triatomic releases m; $|m| = \Lambda$ opens at order $s^{2\Lambda}$, the opening order and angular factors are exact rationals and the slopes are exponentials in R (Paper 32's multicentre-composition remark). Renner and Walsh are the prior art; the corpus's record says "not novel as a method."

- **J6, water's three-centre integrals (W).** The shared-centre (Barnett–Coulson) channels converge to about 10^{-9} by $L \approx 12$; a closed form meets J3's wall (Paper 59, scope section).

- **J7, a three-electron chain (F for the K3 fibre and its elliptic-squared period; the content beyond them is evidence only).** The integral $\langle \rho_a \rho_b \rho_c / (r_{12} r_{13}) \rangle$ over two-centre densities has two momentum integrations and three roots; its fibre is a K3 surface (the double cover of $\mathbb{P}^2$ branched on six lines), the holomorphic period reduces to a three-scale Bessel moment $\int K_0 K_0 K_0$, a sum of two products $K\,K'$ — "elliptic squared" — and the one-centre limit is rational. At equal scales the true fibre collapses to K3 periods and π^3 (evidence); at four generic scale triples, searches in five bases that add dilogarithms, trilogarithms and products of elliptic periods find no relation, nor at two threshold triples (evidence, not proof); on a one-parameter slice the fibre is a K3 function integrated once (Paper 63 `sec:density_ring`). The corpus's contribution is the reduction from a Slater chain to that moment, not the closed form, which Ref. [27] obtains from an identity of Bailey [26]. Pre-registered on 2026-10-13; five falsifiers passed; home Paper 63 [24] (`sec:three_electron`), guard `tests/test_paper63_three_electron_k3.py`.

- **J8, the four-body r_{12} bridge (Obs, validated reduction).** The only four-electron-connected term of the explicit-r_{12} energy, $f_{12} f_{34} / r_{13}$, reduces without a resolution of the identity to a finite contraction of three-body vertex kernels and, for s leaves, to a two-electron Slater–Condon integral (Paper 12; validated against Monte Carlo, no derivation, $N \geq 5$ conjectural). Over two-centre densities the unreduced bridge is a four-density tree, so its holomorphic period sits on the four-density rung of J10, through J10's assumed three-dimensional reduction; the reduction to geminal-dressed densities, validated rather than derived, changes the cost, not the value. The one-centre reduction is classical Hylleraas-CI (King; Sims and Hagstrom); only the two-centre extension is the corpus's.

- **J9, the rung rule (Obs).** For a two-electron integral with a rational kernel the transcendence class is set by the number of distinct two-centre densities the kernel couples — zero or one gives genus zero, two gives genus one, while two densities on the same centre pair (the two-centre exchange class) close in the two-centre seed set although they build the same fibre family, so the fibre genus alone does not decide the class — and not by the number of nuclei (Paper 59, geminal section, symbolic at one centre and test-backed at several); three densities over two momenta give a surface rather than a higher-genus curve (J7), and four give the first rung not built from elliptic periods (J10). The ordering holds for trees: a Coulomb ring of three densities sits on the elliptic rung, and the correlated triangle with exponential geminals on the K3 rung at $\gamma = 1$ (J11). The same-pair clause belongs to the Coulomb kernel: with a Slater geminal the two-centre master integral's operator is genus one, and the kernel's mass and shape move the correlated triangle's class (Paper 59). The rule is a pattern across the tested instances, not a derivation.

- **J10, four linked densities (F for the tree identity and the non-elliptic operator).** The holomorphic period of any tree of two-centre densities depends only on how many densities it links; for four, the one-scale family satisfies an irreducible fourth-order Calabi–Yau-type operator with no quadratic relation among its solutions and a non-constant Yukawa coupling, so it is not built from elliptic periods. The operator is entry #58 of the Calabi–Yau tables, a Hadamard product; the corpus's contribution is the route from a molecular integral to it. Five densities give no new type, and for two to five densities the tree ladder is the banana ladder of Feynman integrals under a Hadamard product with the Legendre family, so those operators are prior art; and the reduction of the real three-dimensional integral into the one-dimensional moment family is shown for both trees of four, the star and the path, each with a piece equivalent to that operator and no K3 piece. Pre-registered on 2026-10-16; home Paper 63 [24] (`sec:four_density`), guard `tests/test_paper63_four_density_rung.py`.

- **J11, a ring of three densities (F for the loop reduction, the operator and its**

factorization). The Coulomb triangle joins every pair of three densities by a kernel. Its loop momentum integrates out exactly by the three-dimensional uniqueness (star-triangle) relation, leaving the chain's two momenta and three densities. At the fingerprint fibre its operator factors exactly into elementary, algebraic and elliptic pieces, the last equivalent to the Legendre equation of the two-density period, so the ring sits on the elliptic rung, below the three-density chain's K3; at one centre it carries dilogarithms, known from Remiddi's closed form (Paper 63 [24]), where the chain is rational. The registered decision rule did not anticipate a landing below the chain. The drop is read as a property of uniqueness, not of rings (an Observation): the correlated triangle with exponential geminals on two edges sits with the chain on the K3 rung at $\gamma = 1$, by an exact factorization and map with the operator checked on the function numerically, and stays on that rung at $\gamma = 3/2$, where an exact map ties its order-three piece to F_3's operator, no piece sits above it, and its order-two piece is the period of a rational elliptic surface, the operator of a two-dimensional bubble integrated over a line of external momentum moved off the origin by $i\gamma$, a reading that held at $\gamma = 2$ when registered before computing; this falsified its own registered prediction. These placements rest partly on measurement and are outside the F. Pre-registered and committed on 2026-10-17 before the computation; home Paper 63 [24] (`sec:density_ring`), guard `tests/test_paper63_density_ring.py`.

- **J12, a spin grown large (F for the surviving label and the contraction).** As S grows the spin algebra contracts to the oscillator algebra (Holstein–Primakoff; Inönü–Wigner): $n = S - m$ survives as the oscillator quantum number, $[A_S, A_S^\dagger] = 1 - n/S$ exactly, and only the cap $n \leq 2S$ is released. At finite S the label side carries no π; $\pi^{1/2}$ and e^{-x^2} enter only as limits of rational label data, and the sphere's coherent-state measure contracts to the Bargmann measure, so the Bargmann π is the normalization by $\mathrm{Vol}(S^2)$, an M1 reading at Paper 18's tier. The density corrections and eigenvector rates are measured and outside the F. Pre-registered and committed on 2026-10-18 before the computation; home Paper 24 (`sec:large_spin`), guard `tests/test_paper24_large_spin_door.py`.

- **J13, the hydrogen continuum (F).** Continuing $n \to i\eta$, $\eta = Z/k$, carries Fock's S^3 onto Bander and Itzykson's hyperboloid; l, m and the Gaunt rules survive, and only the cap $l \leq n - 1$ is released. The energy-normalized continuum radial density is the continued bound density per unit energy times one factor, $1/(1 - e^{-2\pi\eta})$, at every r and l: the ratio of the two normalizations. That exponential of π is the whole continuum price beyond the continued skeleton, flat at threshold, and a new class rather than a rung of the period ladder. Pre-registered and committed on 2026-10-18 before the computation; home Paper 7 (`sec:continuum`), guard `tests/test_paper7_hydrogen_continuum_door.py`. Its third shape, zero energy, is flat space: the Runge–Lenz algebra contracts to the Euclidean group, and both sides meet one π-free threshold function (registered and guarded the same day, `tests/test_paper7_zero_energy_door.py`).

- **J14, a finite nucleus (C).** A nucleus of finite size is a second length scale inside the electron's, so the rigidity of the S^3 projection breaks. The labels l, m, the Gaunt rules and the node count survive; n is released from being a whole number, with a quantum defect $-(2/5)(ZR)^2$ for every s level at leading order, the Foldy–Friar term read on the labels, then the Friar moment; from fourth order a logarithm of ZR, with γ_E and $\ln 2$ in a profile-dependent constant beside it, and a dependence on n (Paper 23, `sec:finite_nucleus`). The defect continues across threshold as one function of energy, the continuum adding only J13's factor. The radius, then the Friar moment, then $\langle r^4 \rangle$ with a log-moment constant are consumed as matching coefficients, and no label fixes them: the first released axis whose price is calibration data, at multi-focal depth two, and the seventh instance of the multi-focal composition wall (§4). Pre-registered and commit-

ted on 2026-10-18 before the computation; home Paper 23 (`sec:finite_nucleus`), guard `tests/test_paper23_finite_nucleus_door.py`.

- **J15, the Kustaanheimo–Stiefel lift (F).** Lifting $\mathbb{R}^3$ to $\mathbb{C}^2$ by the Hopf map releases one dimension, the $U(1)$ fibre, kept as a constraint. At fixed energy the Coulomb problem becomes a four-dimensional isotropic oscillator, the energy consumed as the frequency and the charge coming out as the eigenvalue. The fibre-neutral sector holds the hydrogen label set with degeneracy n^2 and the Casimir $n^2 - 1$ on the oscillator's integer lattice, and each state pushed down is a Coulomb Sturmian with $Z = np_0$: a linear projection with one universal constant. The price is one fibre volume, $\pi/4$, which cancels in every ratio. Since the oscillator's spectral zeta values carry π^2 at even argument as those of S^3 do, the calibration-π layer of Paper 24's Coulomb/HO asymmetry is withdrawn. The lift is Kustaanheimo and Stiefel's; the contribution is the accounting. Pre-registered and committed on 2026-10-19 before the computation; home Paper 24 (`sec:ks_frame`), guard `tests/test_paper24_ks_lift.py`.

- **J16, a finite nuclear magnetization (C).** The hyperfine factor of an s level adds a third length scale. The labels survive as in J14; at leading order the factor changes by $-2Zr_Z$, with r_Z the Zemach radius of charge and magnetization together, the same for every n, and a logarithm $\frac{4}{5}(ZR)^2 \ln(ZR)$ follows, -2 times J14's leading defect, set by the charge alone; a point charge brings no logarithm. The joint moments are consumed as matching coefficients (Karshenboim's $\psi(n)$ form and logarithm are prior art), so the multi-focal instance HF-4 (§4) is priced like J14. The registered number-class prediction failed in part, on an unregistered $(ZR)^3 \ln(ZR)$ term. Zemach's term is prior art. Pre-registered and committed on 2026-10-19 before the computation; home Paper 23 (`sec:finite_nucleus`), guard `tests/test_paper23_hyperfine_door.py`.

The J rows are indexed by door, with their frozen falsifiers, in `docs/atlas.md`, which `tests/test_atlas_register.py` gates against the catalogue, the owning paper labels and the test files.

2.8 Foundations of the operator-algebraic structure (within C)

The math.OA arc proves WH1's commutative-factor convergence (Paper 38; the almost-commutative extension remains open) via the unconditional lifted-state theorem (Paper 38, main theorem: a Fejér-kernel bound whose constant is $4/\pi$, [INTERNAL THEOREM + MEASURED], and a measured $O(1/n)$ with a tapered amplitude) [15]. Paper 40 extends the convergence to compact connected semisimple Lie groups (for general G conditional on the per-group spin-window decomposition) and derives the Fejér-moment constant at rank 1, leaving its group-independence open [16]. Paper 42 § 11 (the Krein-carrier extension, formerly Paper 43) records the Hilbert–Schmidt orthogonality $\langle H_{\text{local}}, D_W^L \rangle_{\text{HS}} = 0$ at bit-exact precision as a chirality-parity identity independent of β [17]; the $1/\pi^2$ in its residual is the square of the posited normalisation $\beta = 2\pi$ (the temporal-compactification π of Paper 35), not an output of the master Mellin engine.

2.9 Catalogue

A consolidated table of the 44 forced entries (plus the four Observation rows and the three walls) with mechanism, witness, and the three discriminator columns is given in the companion catalogue `docs/forced_free_seam.md`. The structural pattern across the forced inventory is: every entry but four sits at multi-focal depth MF = 1 (the exceptions are C9, the two-loop $S_{\min}$ period, and the rows with more than one momentum integration: J7, the three-electron chain; J10, four linked densities; and J11, the ring), the period-class assignment is either "none"

(skeleton / gauge / counting facts), "meta" (the engine itself), $M_1/M_2/M_3$ (specific period-content forced facts), or — on the released-axis rows — weight-1 / elliptic / K3 / Calabi–Yau type (classes the engine does not reach), and dimensional character is overwhelmingly dimensionless — the gravity sub-domain D2 (S_{BH}, G_N) being the prominent exception, which is itself only dimensionful by virtue of carrying the external cutoff scale Λ.

3 The free side

We organize the free inventory along the same sub-domain axes. The decisive distinction in this section is between two structurally different families of calibration content, an observation that emerges from the catalogue itself rather than being imposed on it.

3.1 Standard Model inner factor (F)

The most load-bearing free row is the Standard Model inner factor. Sprint H1's Yukawa non-selection theorem [11] establishes structurally that no inner fluctuation of D_F on $\mathcal{A}_F = \mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ autonomously selects the Yukawa values $\{y_e, y_\mu, y_\tau, y_u, y_d, y_s, y_c, y_b, y_t\}$. The corresponding empirical confirmation is the 162-cell PSLQ sweep of Sprint Yukawa-PSLQ [32]: at coefficient ceiling $M \leq 1000$, at both the M_Z and the $\mu = 2 \times 10^{16}$ GeV unification scales, in three transforms (y_f, y_f^2, $\log y_f$), with the basis sharpened to $M_1 \cup M_2$ (pure-Tate; Paper 18 §IV.6 `thm:eta_trivialization` removes the M_3 slot on Connes–Chamseddine-compatible inner factors, while the Yukawa values carry their own generators y_i, Paper 32), *zero hits* were returned. The clean negative across 162 cells is the strongest available calibration verdict at sprint scale.

The generation count $N_{\mathrm{gen}} = 3$ is similarly free. Sprint C3 (2026-06-08, `debug/sprint_c3_n_gen_non_selec` upgraded the prior Direction 2 + Read 2 NO-GO scopings to a proof-by-argument structural result: the N_{gen} non-selection theorem (Paper 32 `thm:n_gen_non_selection`) states that under the canonical Chamseddine–Connes SM representation, for every integer $N \geq 1$ the GeoVac spectral triple with inner Hilbert-space multiplicity N satisfies every axiom in the framework's current set $\mathcal{A}$, hence $\mathcal{A}$ contains no morphism selecting a specific value of N. The conditional on canonical representation is honest; the unconditional question (whether any SM-phenomenologically-consistent rep correlates N with algebra factor count under an alternative rep) remains the open long-range Read 2 sharpest-falsifier target. The Direction 2 NO-GO and Read 2 NO-GO [33, 34] supply the structural ingredients: (a) the current Paper 0 packing axiom does not reach the multiplicity-three structure as a consequence (Direction 2), and (b) the candidate Hopf-tower-to-representation shortcut fails on three independent grounds in the standard CCM representation (Read 2) — the 32 fermion DOF per generation are distributed across all three algebra factors, the CKM/PMNS cross-generation Yukawa coupling is structurally required, and the DOF count is wrong by a factor of 8 in the alternative reading. No published proposal in the spectral-triple literature (Krajewski 1998, Connes 2006, CCM 2007, Devastato–Lizzi 2013–2015, Boyle–Farnsworth 2018, Ma 2018, Bochniak–Sitarz 2025, Hessam–Khalkhali) derives $N_{\mathrm{gen}} = 3$ from a Hopf-tower argument; Ma 2018 derives it via an independent algebra-extension axiom, not from the existing Paper 0 input.

The inner-factor KO-dimensional signature is similarly free. Sprint F3 (2026-06-08, `debug/sprint_f3_ko_di` upgrades the prior Direction 2 packing-reach NO-GO scoping into a proof-by-argument structural result: the inner KO-dim non-selection theorem (Paper 32 `thm:ko_dim_non_selection`) composes Paper 0 §VII.B's explicit kinematic-scope statement (packing produces labels and graph topology, but no Dirac operator, no real structure, no chirality grading) with the Door 4f / Door 4b finding that inner KO-dim is a property of (D_F, J_F, γ_F). The composition is one line: $\mathcal{A}$ produces no (D_F, J_F, γ_F) data autonomously, hence cannot autonomously select KO-dim. The conditional is again the canonical CCM rep; the unconditional version (whether any non-

canonical SM-phenomenologically-consistent rep yields a different KO-dim) remains open. The CKM and PMNS mixing-angle entries inherit F1 + F4 structurally (Boyle–Farnsworth gives the shape but not the values). The Higgs VEV v and the neutrino mass values are dimensionful scales consumed externally.

3.2 Multi-focal-composition walls (G; cf. §4)

Six independent observables (G1–G6: the HF-3 recoil cross-register, the HF-4 Zemach magnetization density, HF-5 multi-loop QED on hyperfine, LS-8a two-loop self-energy renormalization, the W1e chemistry inner-region overattraction, and the H1 Yukawa non-selection) converge on the same structural reason for being free. Each is described once, in §4, where their unity is the subject. All six entries sit at multi-focal depth MF > 1, period class "outside" the master Mellin engine, and dimensional character mostly dimensionful (the exception being the Yukawa values themselves, which are dimensionless but lie outside the $M_1 \cup M_2$ pure-Tate basis tested).

3.3 Gravity calibration (D5–D6)

The Mellin moments $\varphi(0)$, $\varphi(1)$, $\varphi(2)$ of the cutoff function f are externally supplied. The sector-wise moment map (tip $\leftrightarrow \varphi(0)$, Einstein–Hilbert $\leftrightarrow \varphi(1)$, $\Lambda_{cc} \leftrightarrow \varphi(2)$) is proven and the relations among the gravitational observables are forced (D7), but the moment *values* are not. The cosmological-constant problem in GeoVac is structurally precise: obtaining the observed dimensionless ratio $\Lambda_{cc} \cdot G_{\text{eff}} \approx 36\pi \cdot \varphi(2)/\varphi(1)^2 \sim 10^{-124}$ requires a cutoff function with $\varphi(2)/\varphi(1)^2 \approx 10^{-124}$ while $\varphi(0), \varphi(1) = O(1)$, which no natural cutoff produces. This is a calibration selection problem on the cutoff function, structurally similar to the Higgs–Yukawa selection problem but on a different object.

3.4 Chemistry-side inner-factor input data (H6–H7)

The Clementi–Raimondi 1963 atomic-Hartree–Fock fits supply the screening exponents $\zeta_{1s}, \zeta_{2s}, \zeta_{2p}$ used in the FrozenCore $Z_{\text{eff}}(r)$ profile for second-row and heavier chemistry. The multi-zeta valence basis coefficients are similarly external. The W1e period-class diagnostic [35] classifies these as the chemistry-side analog of the SM inner-factor input data (Paper 18 §IV.6 sixth-tier reading): the framework provides the multi-electron Hamiltonian assembly machinery (composed_qubit, balanced_coupled) on top of these inputs but does not autonomously select the inputs. The chemistry analog of η-trivialization (Paper 18 §IV.6, `thm:eta_trivialization`) is the open structural question on this side.

3.5 Foundational calibration (Class 1; I)

Four entries sit at the foundational level. The fine-structure constant α inherits E6 (the combination rule is an Observation; the value is external). The Born rule probability assignment $p = |\langle a|\psi\rangle|^2$ is structurally external: the framework uses Gleason's theorem via Hilbert-space inheritance and does not improve on Gleason. The Higgs direction $\hat{n} \in S^2$ in the Boyle–Farnsworth parameterization [28] is similarly external, with an open structural question of whether $\hat{n}$ should be identified with the Hopf base S^2 of S^3 (whose area $\text{Vol}(S^2)/4 = \pi$ is read, a reading, as the M_1 volume-slot value) — a flagged-but-not-pursued follow-on. The fourth is the temporal observation window β (I4). Once time is compactified to a circle, the 2π per Matsubara mode is a volume-slot reading (Paper 35 [13]); the window itself is observer-supplied and is consumed. Paper 35 reads a finite real-time window as a Euclidean circle of period β, an identification it adopts rather than derives: a compact interval without periodic identification keeps a continuous spectrum, and the Matsubara circle is Euclidean while the window is real-time. Whether a real-time window forces the circle is open.

3.6 Two structurally distinct families on the free side

The catalogue reveals an observation that was anticipated but never sharply tested in the prior literature.

Observation 3.1 (Two-family structure of calibration content)**.** The free-side entries decompose into two structurally distinct families with different discriminator-column signatures:

1. **Multi-focal composition family** (G1–G6, E6, E7, E8, F1, F4, F5, F6, F7, D5, D6, I1; 17 entries): "the framework can build it but cannot autonomously pick a number." Signature: MF > 1, period class outside, mostly dimensionful.

2. **Inner-factor input data family** (F2, F3, B8, H6, H7, I2, I3): "categorical to the framework's outer-factor mechanism; lives elsewhere by structure." Signature: MF $= 1$, period class "none" or N/A, dimensionless.

This is the operative refinement of the structural-skeleton scope pattern. The earlier one-bucket framing ("structural skeleton vs. calibration data") was correct but coarse. The two-family observation aligns with the previously-named external-input three-class partition, with the third class (multi-determinant correlation) appearing only in the chemistry sub-domain as W1e and currently best read as a sub-case of family 1 with chemistry-specific composition machinery. The released-axis rows of §2.7 belong to neither family, with one exception: they are not consumed (J14, a finite nucleus, consumes its radius at two focal lengths and so sits with the multi-focal composition family), the multi-focal depth above one of J7, J10 and J11 comes from momentum integration rather than multi-focal composition, and the three that sit at Observation status (J2, J8, J9) do so for want of a derivation, not for want of a mechanism. They are indexed by door rather than by family in `docs/atlas.md`.

4 Cross-domain unity: the multi-focal-composition wall

The most structurally striking pattern in the catalogue is that six observables across four sub-domains, and a seventh on a released axis, converge on a single structural reason for being free.

4.1 The seven instances

Listed in order of historical surfacing:

1. **H1 (Sprint H1, 2026-05-31; SM-side)**. $\mathcal{A}_F$ admits the Higgs structurally but does not autonomously select Yukawa values.

2. **LS-8a (2026-05; QED two-loop SE)**. Iterated Chamseddine–Connes spectral action reproduces the two-loop integrand structure but cannot generate $Z_2, \delta m$ counterterms.

3. **HF-3 (Sprint HF, 2026-05-07)**. Track NI cross-register couples discrete spin labels but does not couple spatial coordinates between electron and nucleus registers.

4. **HF-4 (Sprint HF, 2026-05-07)**. hyperfine_coupling_pauli is point-like in proton position; no magnetization-density distribution carried. With the distribution carried the instance is priced like J14: the electron's labels survive, and the hyperfine factor is set by the Zemach moment of charge and magnetization together, then further joint constants as matching coefficients (Paper 23 [6], `sec:finite_nucleus`).

5. **HF-5 (Sprint HF, 2026-05-07)**. Bare two-loop vertex topologies diverge $\sim N^{3.79}$ with no autonomous renormalization (cf. LS-8a).

6. **W1e (Sprint W1c arc, 2026-05-23)**. NaH inner-region overattraction; six framework-internal modification attempts fail at sprint scale, and the W1e period-class diagnostic returns zero genuine outer-factor ($M_1/M_2/M_3$) identifications across its eleven correction terms [35] — the chemistry-side analog of the Yukawa non-selection theorem.

7. **J14 (a finite nucleus, released-axis row)**. The electron's labels survive a finite nucleus exactly, while the size of every effect is set by moments of the nuclear charge that no label fixes: the mean square radius, then the Friar moment (Paper 23 [6], `sec:finite_nucleus`). The instance needs no loops and no relativity.

4.2 The unifying statement

Observation 4.1 (Multi-focal-composition wall)**.** The framework couples discrete labels (quantum numbers $n, l, m, m_s, \kappa, m_j$, isospin, color, gauge representations) cleanly via Wigner symbols and selection rules. Single-focal-length QED on S^3 is handled including curvature corrections at structural level (Parker–Toms verified at 0.5% in HF-2). Spatial coordinates exist only after a Fock-style projection fixes a focal length; multi-focal observables that require composing two or more such projections at once (electron–nucleus V_{ne} with quantum-mechanical nucleus position, proton magnetization × electron wavefunction, electron–electron r_{12} where each electron carries its own scale, multi-loop QED renormalization between bare integrand and counterterm structure) lie outside the framework's native composition apparatus.

[PRIOR ART] For the nuclear instances (J14, and HF-4 through J16) the price the wall charges is the matching statement of nonrelativistic QED [7]: a structured nucleus enters only through contact-operator coefficients, and logarithms come from iterating lower-order contacts [8]. Those two instances therefore share one textbook mechanism, and they support the wall's existence, not that it is specific to GeoVac; at the Observation's tier, its GeoVac-specific content there is only that the label skeleton has no native matching step, as Paper 32 says of the counterterm instance LS-8a.

The G4b cross-manifold question $\mathcal{T}_{S^3} \otimes \mathcal{T}_{S^5}$ is the spectral-triple-level instance of the same wall. Paper 24 §V records this as the fourth Coulomb / HO asymmetry layer [9].

4.3 Operational consequence

When dispatching new investigations, the catalogue affords a forward-classification:

- **Discrete-label couplings** → framework handles natively. Single sprint, expect positive.
- **Single-focal-length QED on S^3** → framework handles natively; curvature corrections expected (Parker–Toms 0.5%).
- **Multi-focal spatial composition** → framework hits Observation 4.1. Sprint should be diagnostic, naming which composition is missing, not implementation. Expect negative-with-clear-naming.
- **Renormalization counterterms** → framework hits the LS-8a wall (= Observation 4.1 sub-case).

This classification is what permits the catalogue to function as a research-direction tool rather than a static record: candidate investigations can be pre-classified as Class A (handle), Class B (handle, light corrections), Class C (diagnostic only), Class D (out of scope at present axiom level) before resources are committed.

5 Candidate discriminator principles: a preliminary probe

This section probes five candidate principles for what distinguishes forced from free content. None of them is established as *the* principle. They are tested against the catalogue and reported on; the analysis is preliminary by design.

5.1 Candidate P1: multi-focal depth

Principle. Forced iff MF = 1; free iff MF > 1.

Coverage. Every multi-focal-composition entry (G1–G6, E6, E7, E8, F1, F4, F5, F6, F7, D5, D6, I1) sits at MF > 1, correctly placed as free. Every forced entry except C9 (the two-loop $S_{\min}$ period) and the released-axis rows J7, J10 and J11 (more than one momentum integration), all tagged MF > 1, sits at MF = 1.

Misclassifications. On the 60-entry catalogue of Table 1, eight in all. Seven entries break P1 in the forced direction: F2 (N_{gen}), F3 (KO-dim), B8 (gauge lower bound), H6, H7 (chemistry calibration), I2 (Born rule), I3 (Higgs direction) — all MF = 1 but free; the eighth is C9, forced at MF > 1, and the released-axis rows J7, J10 and J11, added since, are three more of that kind. The seven are precisely the entries of the *inner-factor input data* family (Observation 3.1, item 2). P1 catches the multi-focal composition family cleanly but is blind to the inner-factor input data family.

Verdict. P1 is a one-family discriminator (it discriminates the multi-focal composition family) but does not generalize. The empirical evidence is consistent with the two-family structure of Observation 3.1: multi-focal depth is the right axis *for* the multi-focal composition family, full stop.

5.2 Candidate P2: period-content classifiability

Principle. Forced iff the entry is classifiable in the master Mellin engine $M_1/M_2/M_3$ ring; free iff it sits outside.

Coverage. The multi-focal walls sit at "outside," correctly placed as free, except E6 and I1 (the α entries, period class M1+M2). The forced spectral / period entries C1–C8 sit at $M_1/M_2/M_3$ or "meta," correctly placed as forced, C9 included: its period class, tagged "outside" on the 2026-06-08 labelling, is M3 (the "irreducible" reading was a basis-coverage artifact, identified 2026-06-11).

Misclassifications. Seven (Table 1; eight on the 2026-06-08 labelling, which tagged C9 "outside"): E6 and I1 (free but inside the ring, M1+M2), and B8, F2, F3, I2, I3 (free with period class "none" or N/A, so not "outside"). The predicate as computed reads only "outside" as free, so the gauge / counting forced entries (A1–A5, B1–B7), which mostly sit at period class "none," are classified correctly by it, but for the wrong reason: they are forced not because they live in the period ring but because they are non-period combinatorial / representation-theoretic facts, and "period class none" applies equally to the free inner-factor entries F2, F3.

Verdict. P2 catches the multi-focal composition family but is the wrong axis for the rest of the forced inventory and for the inner-factor input data family.

5.3 Candidate P3: dimensional character

Principle. Forced iff dimensionless; free iff dimensionful.

Coverage. Most forced entries (skeleton, gauge, periods, chemistry/QC scaling) are dimensionless ratios. Most multi-focal-wall free entries are dimensionful scales (cutoff f, Λ_{cc}, v, etc.).

Misclassifications. D2 (S_{BH}, G_N) is forced and dimensionful. F1 (Yukawa values), F2 (N_{gen}), F3 (KO-dim), F5 (CKM), F6 (PMNS), B8 (gauge lower bound) are all free and dimen-

sionless. Cell by cell the count is 14 misclassifications (Table 1; the list above is the representative subset).

Verdict. P3 catches a correlation but not a principle. The correlation reflects that dimensionful objects often require a scale-supplying calibration step, but it does not capture the structural source of the free / forced split.

5.4 Candidate P4: compactness inheritance

Principle. Forced iff derivable from the compact / Peter–Weyl regime of the Paper 0 packing axiom; free iff requires continuum (un-compact) data.

Coverage. The skeleton / spectral / Mellin-engine forced entries (A, C) derive from Peter–Weyl on compact S^3, correctly placed as forced. The cutoff function f, the Higgs VEV v, the multi-loop counterterms are continuum-side data, correctly placed as free.

Misclassifications. As stated informally, P4 shares the blind spot of P1 / P2: the inner-factor input data family (F2, F3, H6, H7, I2) is not obviously continuum data, yet it is free, and the boundary between "derivable from Peter–Weyl on compact S^3" and "derivable from compact data, broadly" is not sharp enough to test cell by cell. The refinement used for Table 1 (forced iff packing-reachable and not both multi-focal and period-outside) misclassifies only I3 (and C9 on the 2026-06-08 labelling), but only because it imports packing-reachability, the criterion of P5; its table score is therefore not independent of P5.

Verdict. P4 captures the structural-skeleton scope reading of CLAUDE.md §1.7 WH3 cleanly for the multi-focal composition family (continuum data = un-compact = free), but fails on the inner-factor input data family and needs sharpening to become a formal cell-by-cell predicate.

5.5 Candidate P5: packing-reachability

The four candidates above all fail on the inner-factor input data family. Sprint C2 (2026-06-08, `debug/sprint_c2_principle_hunt_memo.md`) introduces a fifth candidate that does not.

Principle. An entry is *packing-reachable* iff there exists a witness structural derivation from the axiom set

$$\Big\{\text{Paper 0 packing axiom; standard real-spectral-triple axioms (Hermiticity, } J\text{-reality, KO-dim chirality, order-}$$

Hopf-tower truncation (Hurwitz); Bertrand's classical theorem; Upgrade B (sphere-Lie-group axiom adopted);

$$\text{CCM canonical SM representation on } \mathcal{H}_F\Big\}.$$

Predict forced iff packing-reachable.

Coverage. Every forced entry in the catalogue carries a witness in the corpus: Paper 32 §VIII for the Forced-Count Theorem (a derivation), and for the gauge group an organizing reading plus the adopted Upgrade B axiom, not a derivation [11]; Paper 18 for the master Mellin engine [4]; Paper 28 for the QED structural theorems [10]; Paper 38 for GH convergence [15]; Paper 51 for the gravity closed forms [19]; Papers 22, 14, 23 for the chemistry / QC / nuclear scaling structurals [3, 5, 6]. Every calibration entry carries either an explicit non-selection theorem (Sprint H1's Yukawa non-selection; the W1e period-class diagnostic [35]) or a documented no-go / negative-search memo (Direction 2 and Read 2 NO-GOs on N_{gen} and inner KO-dim [33, 34]; LS-8a renormalization wall [14]; 162-cell Yukawa-PSLQ clean negative [32]).

Misclassifications. One entry, I3 (Higgs direction $\hat{n} \in S^2$), is tagged *conditional*: the Boyle–Farnsworth Higgs unit vector could in principle be identified with a GeoVac S^2 (the Hopf base, whose volume enters the M_1 slot only as a reading), an open follow-on flagged but not pursued. Treating conditional as forced gives one false positive (I3 predicted forced, actual calibration). The misclassification is precisely the open identification, not a defect in P5.

Verdict. P5 is the *consolidated statement of the forced/free boundary itself, not an independent discriminator of it.* An entry is tagged packing-reachable exactly when the corpus carries a forcing derivation, and packing-unreachable exactly when it carries a non-selection theorem or documented negative search — but "forcing derivation exists" is the *definition* of Forced and "non-selection theorem / negative search" is the definition of Calibration, so the packing-reachable tag is, by construction, the F/C label under another name. The 98.3% agreement with the F/C status (59/60 on the 2026-06-08 labelling, κ then counted forced; the lone exception I3 is deliberately tagged *conditional* to encode an open identification) is therefore an *internal-consistency check* — it verifies the catalogue applied its own F/C conventions uniformly across 60 entries — not a discovered structural predictor, and we do not present it as one. The genuine discriminator content is carried by multi-focal depth (P1), period class (P2), and the two-family decomposition (Observation 3.1), all tagged independently of the F/C status; P5 organizes that content rather than predicting it (see §6.1).

What P5 unifies. P5 absorbs P1, P2, and P4 as special cases. Family-1 entries fail P5 because the framework lacks a multi-focal composition theorem (MF $> 1 \rightarrow$ no closure $\rightarrow$ packing-unreachable). Family-2 entries fail P5 because their values live in a categorical-external register on which the outer-factor mechanism is structurally silent (η-trivialization on the SM side, Paper 18 §IV.6; chemistry-side analog at the inner-factor input-data tier). The two families exhibit different reasons for failing P5, but both fail it.

5.6 Synthesis after five candidates

Table 1: Candidate principles after Sprint C2. Accuracy is true-positive + true-negative over total entries on the 60-entry catalogue. "Family 1" = multi-focal composition (17 calibration entries); "Family 2" = inner-factor input data (7 calibration entries). P5 is the consolidated F/C boundary (definitionally the status column); its 98.3% is an internal-consistency check, not independent accuracy (§6.1).

Principle	Accuracy	Family 1	Family 2	Misclass.
P1 multi-focal depth	86.7%	✓	misses	8
P2 period classifiable	88.3%	✓	misses	7
P3 dimensional character	76.7%	partial	misses	14
P4 compactness inheritance	98.3%	✓	one (open)	1
P5 packing-reachability	**98.3%**	✓	✓	**1 (open)**
P1 $\wedge$ P5	96.7%	✓	one (open)	2
P1 $\vee$ P5	88.3%	✓	misses	7

P5 (packing-reachability) is the *consolidated restatement* of the forced/free boundary, not an independent fifth discriminator: its 98.3% agreement (computed on the 2026-06-08 labelling, in which κ was counted forced; κ is now an Observation row outside the F/C binary) with the F/C status is an internal-consistency check (the packing-reachable tag is the F/C label by construction; §6.1). The genuine discriminator content is the two-family decomposition and the independently-tagged multi-focal-depth and period-class axes.

Observation 3.1's two-family decomposition is preserved as the structural failure-mode story beneath P5, not as evidence that two separate principles are needed. The two families share the surface verdict (packing-unreachable) but exhibit different structural reasons for it, with η-trivialization on the inner-factor side (family 2) and multi-focal-composition gap on the projection-apparatus side (family 1).

The honest interpretation: the forced/free seam IS the packing-reachable / packing-unreachable boundary, and P5 *names* that boundary rather than independently predicting it. The sharpened

open question — whether the boundary admits a *meta-theorem* that decides forcedness without first locating the witness — is §6.

6 Open questions

6.1 P5 and the tautology question (resolved)

Sprint C2 (§5.5) introduced packing-reachability and left open whether it is a structural principle or is definitionally equivalent to the F/C label.

The tautology concern. The packing-reachable tag for each entry was determined by examining the corpus for witness derivations (forced entries) and explicit non-selection theorems / negative-search memos (calibration entries). If "packing-reachable" is operationally just "we have a derivation," then P5 = F/C label by construction, and the 98.3% agreement is an internal-consistency check rather than an independent verdict.

A direct check settles it. We attempted to determine packing-reachability by an *F/C-blind* criterion — "does the corpus carry a cited theorem or bit-exact test for this entry?" — applied identically to every entry. It does not discriminate: all 60 entries carry such a witness (the forced entries via forcing derivations, the calibration entries via the non-selection theorems of Paper 32 §VIII and the negative-search memos), so an F/C-blind witness-existence check predicts "forced" for all 60 and scores at the 60% base rate (36/60 on the 2026-06-08 labelling: the forced entries plus the admitted one, the audit's convention; 34/60 = 56.7% counting today's forced entries among the same 60). The 98.3% is recovered only by reading each witness's *polarity* (forcing vs. non-selection), and that polarity reading *is* the F/C judgment. We therefore conclude that P5 **cannot be made non-circular**: it is the forced/free boundary restated, and its agreement with the F/C label is an internal-consistency check. This does not weaken the catalogue — the structural discovery is the two-family decomposition and its independent multi-focal-depth / period-class signatures (§5), which P5 consolidates.

The sharpened open question. The open question is correspondingly sharpened: not "is P5 tautological" (it is), but whether the packing-reachable boundary admits a *meta-theorem* — a structural characterization deciding forcedness *without* first locating the forcing-or-non-selection witness. A theorem of the form "any observable on the GeoVac spectral triple is forced iff packing-reachable" would lift the boundary from internal bookkeeping to structural truth. It is long-range, NCG-research-grade, and named here as the sharpest follow-on of the Sprint C2 finding. A first answer — covering the skeleton, the inner-factor family, the admitted value, and, at mechanism level on both corpus-faithful instances, the multi-focal family — is the Bohr-site dichotomy of §6.2.

6.2 A Bohr-site partial meta-theorem (Sprint Topos-1)

For a finite-dimensional C*-algebra A, let $\mathcal{C}(A)$ denote the poset of commutative unital *-subalgebras (the *Bohr site* of the topos approach to quantum logic). Sprint Topos-1 tested the candidate meta-theorem: an entry is **forced** iff it is a property of the site (plus the distinguished GeoVac flag) requiring no valuation data; **calibration** iff it is valuation data — a point the site cannot supply; **admitted** iff the site is *degenerate* (cannot distinguish the candidates); and the *dimension* of each valuation freedom is itself a site-side fact. Four computed legs (backing: `tests/test_topos1_site_invariants.py`, bit-exact):

1. **Site strata** (INTERNAL THEOREM, combinatorial): conjugacy strata of commutative unital *-subalgebras of $M_k(\mathbb{C})$ correspond to integer partitions of k with stratum family

dimension $k^2 - \sum_i \lambda_i^2$: M_2: $\{0, 2\}$; M_3: $\{0, 4, 6\}$; $\mathbb{H}$ (real form): $\{0, 2\}$ (the $\mathbb{C}_q$ family over $S^2/\{\pm\}$).

2. **The B5 correspondence** (MEASURED, order-invariant level): $\mathcal{C}(M_2(\mathbb{C}))$ is the height-1 flat fan (one point stratum plus one 2-parameter family of maximal 2-dimensional subalgebras, moduli $\mathbb{R}P^2$, with no intermediate elements) — the minimal non-rigid Boolean block, which is *precisely* the excluded case of the Döring–Harding / Hamhalter reconstruction theorems [29, 30]: an order isomorphism $\mathcal{C}(A) \to \mathcal{C}(B)$ is induced by a Jordan *-isomorphism for all A with *no type I_2 summand* (and $A \neq \mathbb{C} \oplus \mathbb{C}$), and type I_2 is exactly $M_2(\mathbb{C})$. At $k = 3$ the appearance of intermediate subalgebras (height 2) restores this rigidity. We observe that $\mathcal{C}(\mathbb{H})$, for the real *-algebra $\mathbb{H}$ whose complexification is $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{C} \cong M_2(\mathbb{C})$, shares this height-1 fan and the same $\mathbb{R}P^2$ maximal-element moduli, so the two candidates at the catalogue's lone admitted-not-forced entry (B5, $\mathbb{H}$ vs $M_2(\mathbb{C})$, Door 4c) are indistinguishable by these order invariants; we record this $M_2(\mathbb{C})/\mathbb{H}$ coincidence as a framework observation (not located in the Bohrification literature, which treats complex C*-algebras). We do *not* claim the poset fails to determine $M_2(\mathbb{C})$ as a complex C*-algebra — for finite-dimensional complex A, $\mathcal{C}(A)$ determines A up to *-isomorphism with no exception [31] — only that the Jordan-induction / automorphism-rigidity reconstruction degenerates at type I_2, which is exactly the $k = 2$ slot where the catalogue admits $\mathbb{H}$ over $M_2(\mathbb{C})$. This is decided from site properties without consulting any witness derivation.

3. **Machine-verified Kochen–Specker witness** (bit-exact): the orthogonality hypergraph of primitive integer rays in $\mathbb{R}^3$ with entries in $\{-2, \ldots, 2\}$ (49 rays, 26 orthonormal triples; containing the Conway–Kochen ray family) admits *no* 0/1 frame-function coloring (exhaustive search), while the range-1 set (13 rays) is colorable. Hence dim-≥ 3 blocks are valuation-*obstructed* (their values are necessarily external data), dim-2 blocks are valuation-*choiceable*, dim-1 blocks are pointed — three site statuses matching forced / admitted / calibration structure at the block level.

4. **The GeoVac flag** (bit-exact, $n_{\max} = 2, 3$): the Casimir chain $\langle n \rangle \subset \langle n, L^2 \rangle \subset \langle n, L^2, L_z \rangle$ is a maximal chain of commutative subalgebras of dimensions $(2, 3, 5)$ and $(3, 6, 14)$; the graph adjacency satisfies $[A, L^2] = 0$ exactly (the edge rule conserves l — a selection rule that is itself a site fact) with $\|[A, n]\|, \|[A, L_z]\| > 0$.

5. **Family-1 mechanism: frame-meet collapse** (bit-exact, Sprint Topos-2): for two same-center hydrogenic Fock frames at exponents $Z = 1$ and $Z' \in \{2, 3\}$ (the mixed-exponent setting of the Track-DF / Löwdin wall), the exact rational inter-frame overlap is (l, m)-block-diagonal with *connected* support in every block ($n_{\max} \leq 4$), so the meet of the two frame MASAs is *exactly the angular-grading algebra* (collapse $14 \to 9$ and $30 \to 16$): frame composition preserves the angular structure and nothing else — the site statement of the angular-sparsity / radial-wall seam (Papers 22, 32). The join generates the full matrix algebra on each (l, m) block (connected support + irreducibility), and blocks of dimension ≥ 3 appear at $n_{\max} \geq 3$, so composed radial observables land in Kochen–Specker-obstructed algebras: **Family-1 externality = meet-collapse followed by join-obstruction** — a provably different *route* to valuation-need than Family-2's ab-initio noncommutativity, with the same terminus (matching the two-family decomposition: same surface verdict, different structural reasons). By-product exact lemma: rate coincidence $Z_1/n_1 = Z_2/n_2$ with $|n_1 - n_2| \geq 2$ forces a vanishing inter-frame overlap (same-variable Laguerre band structure; verified exhaustively for $Z' \leq 5$, $n \leq 6$), while the converse is panel-scoped (one sporadic off-coincidence zero at $Z' = 2$, $l = 3$, $n = n' = 5$). Backing: `tests/test_topos2_family1_meetjoin.py`.

6. **Two-center meet: the m-grading exactly** (exact, Sprint Topos-3): for frames at centers separated by R along z ($Z' \in \{1, 3\}$, $R \in \{1, 2\}$, $n_{\max} = 3$), every inter-frame overlap

reduces in prolate spheroidal coordinates to $e^{-p}(Ue^{q} + Ve^{-q})$ with U, V exact rationals (Mulliken auxiliary integrals A_n/B_n; zero-decidability by the linear independence of $e^{\pm q}$ over $\mathbb{Q}$; machinery certified against the classical $\langle 1s|1s\rangle(R) = (1 + R + R^2/3)\, e^{-R}$ as an exact rational identity at $R = 1, 2, 7/2$). Every m-block has *full* exact support — the same-center rate-coincidence zeros do not survive displacement — so the two-center meet is *exactly the m-grading algebra*, confirming the prediction pre-registered above. The resulting **meet-dimension ladder** $14 \to 9 \to 5$ (identical frames → same-center different focal → two-center, $n_{\max} = 3$) measures the symmetry of the composition, and the two-center join is *more* Kochen–Specker-obstructed (blocks of dimension 6 and 3) than the same-center case (dimension 3 only) — the site restatement of cross-center composition being the framework's hardest wall. Backing: `tests/test_topos3_exact_meet.py`.

7. **General-parameter vanishing lemma** (exact, Sprint Topos-4): the by-product lemma of item 5 is closed in general form. For same-l inter-frame overlaps the exact unnormalized overlap factors as $S = (\text{positive prefactor}) \cdot P_{n_1,n_2,l}(t)$ in the rate variable $t = (Z_1/n_1 - Z_2/n_2)/(Z_1/n_1 + Z_2/n_2) \in (-1, 1)$, with P an explicit polynomial of degree $D = n_1 + n_2 - 2l - 2$ (for $n_1 = n_2$, P is even in t by frame-swap symmetry; for the single-radial-node family $n_1 = n_2 = l + 2$, where $D = 2$, it is exactly the symmetric Jacobi / Gegenbauer polynomial $P_2^{(l,l)}$, but for higher D it is not in general a classical Jacobi), so $S = 0 \Leftrightarrow P(t) = 0$. The roots fall into three structural families: (i) $t = 0$ (rate coincidence), a root *exactly* when $|n_1 - n_2| \geq 2$ (proved over the n, l grid — the structural form of item 5's exhaustively-verified ($\Leftarrow$) direction); (ii) $t = (n_2 - n_1)/(n_2 + n_1)$ (equal-exponent hydrogenic orthogonality $\langle n_1 l|n_2 l\rangle = 0$), always a root for $n_1 \neq n_2$; and (iii) residual roots, generically *irrational*. The "sporadic" off-coincidence zeros are exactly the *rational* residual roots: in the scan $Z, n < 7$ there are precisely two families, $(n_1, n_2, l) = (5, 5, 3)$ with $P = 1 - 9t^2$ (roots $\pm 1/3$, the item 5 boundary case, realised at $(Z, Z') = (1, 2)$) and $(4, 5, 2)$ with roots $\pm 1/2$; the integer zero census fully accounts $253 = 37 + 210 + 6$ over the three families. Individual-element zeros do not disturb the block *connectivity* on which items 5–6 rest. Backing: `tests/test_topos4_vanishing_lemma.py`.

Under this dichotomy the four proof-by-argument inner-factor results of §6.3 re-read as site statements (the moduli *dimension* (32 full-axiom, 16 matter-sector projection) is a site-side fact; the moduli *point*, the multiplicity factor, and the real-structure signature are valuations), and the lone P5 exception I3 factorises: the Higgs direction splits into an internal *space* (the Hopf-base S^2, read as an M_1 site object) and an external *point* (the valuation $\hat{n}$) — the "conditional" tag is precisely the open question whether the internal space is the GeoVac S^2.

The Family-1 census (Sprint Topos-4). Mapping all 17 multi-focal (MF > 1) calibration entries onto the site mechanism shows that *Family 1 is not mechanism-homogeneous.* The meet-collapse → join-obstruction mechanism (items 5–6) is the Bohr-site image of the *spatial-composition sub-sector specifically* (G1, G2, G5; Paper 32 Theorem `thm:spatial_composition_radial_wall`) — only 3 of the 17 entries. The other 14 reach externality by *valuation, not composition*: six via a test-function / cutoff valuation (E7, E8, G3, G4 renormalization; D5, D6 cutoff moments; Theorems `thm:single_cutoff_spectral_action`, `thm:cutoff_function_external`), and eight via a point / scale / period valuation (F1 Yukawa and its duplicate catalogue label G6; F4, F5, F6, F7; E6, I1). The MF > 1 tag that defines Family 1 is therefore orthogonal to the site-mechanism split: at the site level the two valuation routes carry the *same* status as Family 2 (valuation the site cannot supply), so the operative site taxonomy is **meet-collapse** (spatial-frame composition) **vs. valuation** (everything else, Family 2 included). This sharpens item 5's headline from "Family-1 externality = meet-collapse" (an over-generalisation from the two computed spatial instances) to its precise form: spatial-composition-sub-sector externality = meet-collapse.

Honest scope. The dichotomy is a *partial* meta-theorem: it covers the skeleton (commutative $\mathcal{A}_{\mathrm{GV}}$, Bohr topos Boolean), the inner-factor family (Family 2, via the Kochen–Specker obstruction), the admitted value (site degeneracy), the valuation-freedom dimensions, and — at mechanism level on the three spatial-composition entries — the meet-collapse route of the multi-focal family, all decided from site properties without locating witnesses (the non-circularity P5 could not achieve). The general-parameter vanishing lemma (item 7) and the 17-entry census (above) are now closed; the relation to general site-reconstruction theorems at the exceptional dimension $k = 2$ is stated affirmatively in item 2 above (Döring–Harding / Hamhalter type-I_2 exception [29, 30], with the non-degeneracy of the complex isomorphism class kept explicit [31]). The full 60-entry classification (upgrading the F/C column to a site-derived column) and internal-logic statements of the seven non-selection theorems (and the E6 Observation) remain open catalogue follow-ons. Nothing here touches the Observation status of the combination rule K: its *value* classifies as an external valuation (route 3 of the census), consistent with and not upgrading that label.

6.3 Sprints C3 + F3 closure: four proof-by-argument inner-factor results

Sprints C3 + F3 (2026-06-08, this same close-of-day) upgraded the F2 (N_{gen}) and F3 (inner KO-dim) entries from diagnostic-grade to proof-by-argument grade via Paper 32 `thm:n_gen_non_selection` and `thm:ko_dim_non_selection`. The corpus now carries four proof-by-argument results (the Forced-Count dimension count and three non-selection results) characterising the free-side content of the inner-factor structural-skeleton boundary at the canonical-representation level:

1. **Forced-Count Theorem** (Paper 32 `thm:forced_count`): the moduli dimension $\dim \mathcal{M}(D_F) = 32$ at one generation (matter-sector projection rank 16) is forced, given the admitted $\mathcal{A}_F$ and the imported CCM representation (corrected 2026-09-03 from the degenerate-sample values 128/260; Paper 32).

2. **H1 Yukawa non-selection theorem** (Paper 32 §VIII.C): no inner fluctuation of D_F autonomously selects the Yukawa values within $\mathcal{M}(D_F)$; the diagonal slice carries 8 free parameters per generation.

3. **N_{gen} non-selection theorem** (Paper 32 `thm:n_gen_non_selection`, Sprint C3): no axiom in the framework's current set selects a specific N_{gen}; the multiplicity factor is independent of the axiom set under canonical representation.

4. **Inner KO-dim non-selection theorem** (Paper 32 `thm:ko_dim_non_selection`, Sprint F3): the framework's axiom set is kinematic (Paper 0 §VII.B) and KO-dim is real-structure data; $\mathcal{A}$ cannot autonomously select an inner KO-dim signature.

Together, these four proof-by-argument results characterise the full free-side content of the inner-factor structural-skeleton boundary at the canonical-representation level (Paper 32 `rem:full_inner_facto`: the forced data is the moduli space and its dimension (Forced-Count); the free data is (i) the point in the moduli space (Yukawa values, H1 non-selection), (ii) the multiplicity factor (N_{gen}, C3 non-selection), and (iii) the real-structure signature (inner KO-dim, F3 non-selection). All three free data are calibration inputs in the structural-skeleton-scope sense.

The remaining empirical-only non-selection entries (E6 combination rule K; D5–D6 CC fine-tuning; F4–F7 Higgs VEV / CKM / PMNS / neutrino masses; H6–H7 chemistry inner-factor data; I1–I3 foundational calibration) remain candidates for future upgrades, but they sit outside the inner-factor sector that the four results above now bound.

Sprint E7/E8 closure: multi-loop QED renormalization sub-sector. A fifth item, an Observation rather than a proof-by-argument result, was added in Sprint E7/E8 (also 2026-06-08), in the spectral-action sector. Paper 32 `thm:single_cutoff_spectral_action` records that the Chamseddine–Connes spectral action $S(D, \Lambda) = \mathrm{Tr}\, f(D/\Lambda)$ in the framework's axiom set $\mathcal{A}$ is single-cutoff by stipulation, and that none of the constructions inspected introduces a second scale. The corresponding corollary (`cor:multi_loop_renormalization_wall`, **[CONDITIONAL]** on the unargued premise that renormalization needs a second independent scale; an on-shell scheme is a candidate counterexample) classifies catalogue entries E7 (two-loop SE counterterms $Z_2, \delta m$), E8 (multi-loop QED beyond LS-7), G3 (HF-5 multi-loop hyperfine), and G4 (LS-8a, = E7) as multi-cutoff renormalization calibration data. It offers a conditional structural reading of the LS-8a wall observed empirically in Paper 36 LS-8a (May 2026): the framework reproduces the bare two-loop integrand but has not generated the renormalization counterterms.

Sprint G1/G2/G5 closure: spatial-composition sub-sector. A sixth item, a proof-by-argument non-selection result (with an empirical leg), was added in Sprint G1/G2/G5 (same day, 2026-06-08), closing the remaining spatial-composition sub-sector of the multi-focal-composition wall. Paper 32 `thm:spatial_composition_radial_wall` states that in the framework's tensor-product spectral triple $\mathcal{T}_a \otimes \mathcal{T}_b$, the angular selection rules of composed two-body operators are forced (Paper 54: Gaunt compatibility and m-conservation, structurally; the interaction is pure $k = 0$ monopole at $n_{\max} \in \{2, 3\}$, so the $k \geq 1$ multipoles of Coulomb are absent there) but the radial coupling profile is not autonomously matched to specific physical two-body forms. The structural reason is the non-commutativity of the Fock-projection conformal factor (Paper 7 chordal-distance identity) with the tensor-product construction: two focal lengths give two different conformal factors c_a, c_b whose combined product $c_{ab} \neq c_a \cdot c_b$, and the relative-coordinate reduction from the 6-dim product manifold to the 3-dim relative manifold is not in $\mathcal{A}$.

The corresponding corollary (`cor:spatial_composition_wall`) classifies catalogue entries G1 (HF-3 recoil cross-register V_{eN}), G2 (HF-4 Zemach magnetization), G5 (W1e chemistry inner-region) as spatial-composition calibration data: angular selection rules forced, radial content external. Empirical verification comes from Sprint resolvent_two_body (2026-06-01) which tested four resolvent weightings against Coulomb and found Pearson correlations of at most 0.846 ($n_{\max} = 2$) and 0.796 ($n_{\max} = 3$) for the best weighting (values after the 2026-08-29 ERI-evaluator fix), decreasing with $n_{\max}$.

The multi-focal-composition wall is now classified (spatial half by argument, renormalization half conditionally). Paper 32 `rem:multi_focal_wall_fully_characterized` records the classification of all six unique multi-focal-composition wall instances (G6 = F1 is not independent) under three results:

- **H1 instance**: the H1 Yukawa non-selection theorem (Paper 32 §VIII.C).
- **Renormalization sub-sector** (E7, E8, G3, G4): classified by `thm:single_cutoff_spectral_action` (an Observation) and its conditional corollary.
- **Spatial-composition sub-sector** (G1, G2, G5): closed by Theorem `thm:spatial_composition_radial` (proof-by-argument with an empirical leg).

Five proof-by-argument results and one Observation. After a single sprint wave (Sprints C3 + F3 + E7/E8 + G1/G2/G5 closures, all 2026-06-08):

1. Forced-Count Theorem (inner factor moduli dimension)
2. H1 Yukawa non-selection (inner factor diagonal slice point)

3. N_{gen} non-selection (inner factor multiplicity)
4. Inner KO-dim non-selection (inner factor real-structure signature)
5. Single-cutoff spectral action (multi-loop renormalization sub-sector; an Observation, by stipulation)
6. Spatial-composition radial wall (spatial-composition sub-sector)

The first four close the inner-factor structural-skeleton boundary at the canonical-rep level. The next two classify the multi-focal-composition wall family, the renormalization half conditionally.

Sprint C-arc closure (E6 + D5/D6 + chemistry analog). Two additional non-selection results and one named substantial structural-research target were added in the C-arc terminal-state sprint (same day, 2026-06-08, v3.103.0). One of the two (E6) is an Observation, so the corpus carries *six* proof-by-argument results plus two Observations (the single-cutoff spectral action and E6), and one named substantial target:

7. **No-single-mechanism** $K = \pi(B + F - \Delta)$ (Paper 32, Sprint E6; an *Observation*): no single-mechanism derivation of the empirical combination rule for α^{-1} has been found; the structural argument that the three spectral homes live in exclusive Mellin sub-rings does not hold (the rationals B and Δ lie in every sub-ring), so only the empirical record survives: twelve mechanisms eliminated (Phases 4B–4I + Sprint A α-SP + Sprint K-CC). Combination rule preserved as an Observation per CLAUDE.md §13.5.
8. **Cutoff function f external** (Paper 32 `thm:cutoff_function_external` + `cor:cc_fine_tuning_exter` Sprint D5/D6): cutoff function f and its Mellin moments are external test-function data; no axiom in $\mathcal{A}$ selects moment values. Wald forces relations among moments; values are external. Cosmological-constant fine-tuning $\varphi(2)/\varphi(1)^2 \approx 10^{-124}$ formalised as external moment selection.

Plus one named structural-research target (Paper 32 `rem:chemistry_eta_analog`, Sprint Chemistry-Analog): the chemistry-side η-trivialization analog. Paper 18 `thm:eta_trivialization` restricts the inner factor to the M_1/M_2 Mellin slots via $\{\gamma_F, D_F\} = 0$ (the M_3 slot vanishes); the chemistry-side analog would be a structural property of the FrozenCore $Z_{\text{eff}}(r)$ pipeline mirroring this chirality grading. Sprint-scale path unclear; substantial NCG-research target. Catalogue entries H6, H7 remain empirical-only pending.

The C-arc terminal state. Paper 32 `rem:c_arc_terminal_state` records the full corpus tally: six proof-by-argument results across three sectors (inner-factor / multi-focal-composition / gravity), plus two Observations (the single-cutoff spectral action; E6 in the α combination-rule sector, which holds no theorem), one named substantial follow-on (chemistry η-trivialization analog), all subsidiary residuals (F4–F7 inherit from F1 + F4; I-entries via Gleason / E6 inheritance / open identification) accounted for.

The full forced/free seam is in a stable terminal state for the path-#1 vein C arc. Further sprints on the seam either tackle the named substantial chemistry η-trivialization analog or are residual subsidiaries.

6.4 The chemistry-side η-trivialization analog

A concrete sprint-scale follow-on remains. P5's failure modes match independent structural arguments on the SM side (η-trivialization, Paper 18 §IV.6) but not yet on the chemistry side. The chemistry-side analog would be a structural property of the FrozenCore $Z_{\text{eff}}(r)$ pipeline that mirrors the chirality-grading axiom $\{\gamma_F, D_F\} = 0$ on the SM side. Such a property would give a

categorical reason for the chemistry-side inner-factor input-data entries (H6, H7) to be packing-unreachable. The candidate is well-defined but not currently exhibited; flagged as substantial NCG-research target in the W1e period-class memo [35].

6.5 The "second packing axiom" status

The Direction 2 NO-GO [33] and Read 2 NO-GO [34] establish that the current Paper 0 packing construction does not autonomously reach N_{gen} or the inner KO-dim. The named logical escape is a *sibling axiom* to Paper 0: a new primitive emitting fiber multiplicities, motivated by structural content the existing axiom system does not carry. For a sibling axiom to satisfy the standards of the present catalogue it would need to:

1. Be motivated by structural content the existing axiom system does not carry (not just postulate $N_{\text{gen}} = 3$ directly).
2. Have testable consequences beyond the multiplicity (e.g. predict CKM structure, generation mass hierarchy, or Yukawa–Higgs coupling pattern).
3. Be self-consistent with the Door 4 forced/free seam (i.e. not accidentally also force Yukawa values, which would be inconsistent with the H1 Yukawa non-selection theorem).

These requirements are well-defined but no current construction satisfies all three. The sibling-axiom direction remains the named long-range structural research target.

6.6 Higgs direction and the Hopf base

The Boyle–Farnsworth DGA parameterization [28] represents the Higgs by a unit vector $\hat{n} \in S^2$ specifying the direction of the $\mathbb{C}$ embedding in $\mathbb{H}$. The GeoVac Hopf base S^2 is read (a reading, not a derivation) as carrying the master Mellin engine M_1 volume content $\text{Vol}(S^2)/4 = \pi$. If $\hat{n}$ is identified with the Hopf-base S^2, this would provide a structural correlation between the Higgs orientation and the gauge-bundle base. The follow-on is flagged but not pursued at sprint scale; it sits at the boundary between Observation 3.1 family 2 (where the Higgs direction is currently catalogued) and the multi-focal-composition family.

6.7 Relationship to Connes–Marcolli calibration data in NCG-SM

The free / forced seam catalogued here is consistent with — and arguably a specific instance of — the more general phenomenon visible in the Connes–Marcolli noncommutative-geometric formulation of the Standard Model: the framework admits the gauge group (organizing reading plus an adopted axiom), fixes the factor count and the $\mathbb{C}$, $M_3(\mathbb{C})$ summands under the same reading ($\mathbb{H}$ admitted, not forced), and supplies the spectral action machinery, but the Yukawa values and the generation count enter as external inputs. The GeoVac contribution to this conversation is twofold: the census, organised by the conditional case-exhaustion theorem (first fifteen projections), of which Mellin slots do not appear on the inner factor (the M_3 slot removed via η-trivialization), and the empirical confirmation at 162 PSLQ cells that the SM Yukawas do not land in the $M_1 \cup M_2$ pure-Tate basis at low coefficient ceiling.

The natural follow-on at the cross-community level is engagement with the Brown / Marcolli / Glanois periods program. The two new structural theorems of Paper 18 §IV.6 (η-trivialization and tensor factorization) are paper-level handles for that engagement.

Honest scope

- **Observations-grade content.** This paper consolidates and frames; it does not prove a meta-theorem of the form "any observable on the GeoVac spectral triple is forced iff

packing-reachable." Sprint C2 identifies packing-reachability as a consolidation of the forced/free label whose 98.3% agreement (on the 2026-06-08 labelling) is an internal-consistency check, not an accuracy, and it cannot become an independent discriminator (§6.1); the open question is the meta-theorem above. The two-family structure of Observation 3.1 is empirical, supported by the discriminator-column patterns in the catalogue `docs/forced_free_seam.md`, and is preserved as the failure-mode decomposition beneath P5 rather than as evidence that two separate principles are needed.

- **Catalogue completeness.** The 79 entries (44 forced + 4 Observation + 1 admitted + 27 calibration + 3 walls) are the load-bearing inventory at the time of writing. We may have missed entries; the catalogue's structure permits straightforward addition of new rows as further sprints surface them. The cross-domain unity of §4 is supported by seven instances surfaced across the corpus, two of which (the nuclear ones) share the textbook matching mechanism of nonrelativistic QED, which is past the point of coincidence but is not a closed enumeration.

- **Content used here, not produced.** The Forced-Count Theorem (Paper 32, `thm:forced_count`; a dimension count, proof-by-argument), the case-exhaustion theorem (Paper 32 §VIII; **[CONDITIONAL]**), the master Mellin engine census (Paper 18 §III.7; a reading), the η-trivialization theorem (Paper 18 §IV.6), Paper 22's angular-sparsity theorem, Paper 38's GH-convergence theorem, the Wald two-term-exactness $\Rightarrow$ pure Einstein argument, and the Door 4 series results (partial-door; the seam an Observation-level reading) are all cited; none is reproved here.

- **Empirical-only content used here.** The 162-cell Yukawa-PSLQ clean negative is empirical / numerical at coefficient ceiling $M \leq 10^3$; high-coefficient identities are not ruled out structurally. The W1e period-class diagnostic's 11-term sweep is similarly empirical. The cross-domain unity is seven observed instances, not a theorem.

- **Named open follow-ons (deferred).** The chemistry-side η-trivialization analog (substantial NCG-research target); the sibling-axiom direction toward N_{gen} and inner KO-dim (long-range, no current handle); the Higgs-direction / Hopf-base identification (sprint-scale, NCG-reading-heavy, flagged).

Cross-references

Companion catalogue: `docs/forced_free_seam.md` (this writeup's companion data file, full 79-entry table). Door-indexed view of the released axes: `docs/atlas.md`, gated by `tests/test_atlas_register.py`

Strengthens / consolidates: Paper 32 §VIII (Forced-Count Theorem, Sprint H1 verdict) [11]; Paper 18 §III.7 + §IV.6 (master Mellin engine + inner-factor input-data tier) [4]; Paper 35 (rest-mass vs. temporal-window projections) [13]; Paper 50 (F-theorem ladder, Door 1 forward-run) [18]; Papers 12, 58, 59, 61, 63 (the released-axis rows of §2.7) [20–24]; the prior catalogue `docs/archive/forcing_catalogue.md` (2026-06-01; archived 2026-10-15).

Does not change: WH1's commutative-factor convergence result [15]; WH5 (α as projection constant); the Marcolli–van Suijlekom gauge-network lineage reading of Papers 25, 30, 32.

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[3] J. Loutey, *Paper 14: Qubit Encoding for the Composed Architecture*, GeoVac papers/group4_quantum_computing.

[4] J. Loutey, *Paper 18: Spectral-Geometric Exchange Constants*, GeoVac papers/group3_foundations.

[5] J. Loutey, *Paper 22: Potential-Independent Angular Sparsity*, GeoVac papers/group3_foundations.

[6] J. Loutey, *Paper 23: Nuclear Shell Model Qubit Hamiltonians and Fock Projection Rigidity*, GeoVac papers/group4_quantum_computing.

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[12] J. Loutey, *Paper 33: The 1+6+1 Partition of QED Selection Rules on S^3*, GeoVac papers/group5_qed_gauge.

[13] J. Loutey, *Paper 35: Time as Projection*, GeoVac papers/group6_precision_observations.

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[17] J. Loutey, *Paper 42: Tomita–Takesaki modular structure on truncated $SU(2)$ spectral triples: the four-witness 2π period at finite cutoff*, GeoVac papers/group1_operator_algebras.

[18] J. Loutey, *Paper 50: Boundary CFT_3-on-S^3 Observables and the GeoVac Spectral Triple: Partition Function and Wedge KMS Entropy*, GeoVac papers/group1_operator_algebras.

[19] J. Loutey, *Paper 51: Gravity from the GeoVac Spectral Action*, GeoVac papers/group5_qed_gauge.

[20] J. Loutey, *Paper 12: Algebraic Two-Electron Integrals on the Prolate Spheroidal Lattice*, GeoVac papers/group2_quantum_chemistry.

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[30] J. Hamhalter, *Isomorphisms of ordered structures of abelian C*-subalgebras of C*-algebras*, J. Math. Anal. Appl. **383** (2011) 391–399; doi:10.1016/j.jmaa.2011.05.035. (C*-algebra extension of the reconstruction theorem; uniqueness holds as long as A is not two-dimensional.)

[31] A. J. Lindenhovius, *Classifying finite-dimensional C*-algebras by posets of their commutative C*-subalgebras*, Int. J. Theor. Phys. **54** (2015) 4615–4635; arXiv:1501.03030. (Thm 1: for finite-dimensional A, $\mathcal{C}(A)$ determines A up to *-isomorphism, with no type I_2 exception.)

[32] GeoVac sprint memo: `debug/sprint_yukawa_pslq_memo.md` (2026-06-03): Sprint Yukawa-PSLQ clean negative across 162 PSLQ cells.

[33] GeoVac sprint memo: `debug/seam_packing_scoping_memo.md` (2026-06-03): Direction 2 NO-GO.

[34] GeoVac sprint memo: `debug/sprint_read2_n_gen_scoping_memo.md` (2026-06-03): Read 2 NO-GO on Hopf-tower-to-representation shortcut.

[35] GeoVac sprint memo: `debug/sprint_w1e_period_class_memo.md` (2026-06-04): W1e period-class diagnostic, zero genuine outer-factor identifications.

[36] GeoVac sprint memo: `debug/sprint_c2_principle_hunt_memo.md` (2026-06-08): Sprint C2 principle hunt, packing-reachability at 98.3% agreement (an internal-consistency check) on the 60-entry catalogue.

[37] GeoVac sprint memo: `debug/sprint_c3_n_gen_non_selection_memo.md` (2026-06-08): Sprint C3 N_{gen} non-selection theorem; proof-by-argument upgrade of Direction 2 + Read 2 NO-GO scopings.

[38] GeoVac sprint memo: `debug/sprint_f3_ko_dim_non_selection_memo.md` (2026-06-08): Sprint F3 inner KO-dim non-selection theorem; proof-by-argument upgrade via packing-is-kinematic + KO-dim-is-real-structure-data composition.

[39] GeoVac sprint memo: `debug/sprint_e7_e8_single_cutoff_memo.md` (2026-06-08): Sprint E7/E8 single-cutoff spectral action (an Observation, by stipulation); conditionally classifies the multi-loop QED renormalization entries of the multi-focal-composition wall (E7, E8, G3, G4).

[40] GeoVac sprint memo: `debug/sprint_g1_g2_g5_spatial_composition_memo.md` (2026-06-08): Sprint G1/G2/G5 spatial-composition radial wall theorem; closes spatial-composition sub-sector of multi-focal-composition wall (G1, G2, G5). Multi-focal-composition wall now classified: spatial half by argument at the canonical-rep level, renormalization half conditionally.

[41] GeoVac sprint memo: `debug/sprint_c_arc_closure_e6_d5d6_chemistry_memo.md` (2026-06-08): Sprint C-arc closure; D5/D6 (cutoff function external) theorems + E6 (no-single-mechanism K, now an Observation) + chemistry-side η-trivialization analog scoping; C-arc terminal-state remark tally, now six proof-by-argument results plus two Observations.